Forces on two objects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify an equal-and-opposite interaction pair on different objects.

SUCCESS CRITERIA - I can...

1. Name both objects in an interaction pair.
2. Place the two equal-and-opposite forces on different objects.
3. Explain why equal forces can produce unequal accelerations.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “interaction” and “system” to explain your model or result. Sentence starters are optional.

interaction: two objects affecting one another

system: the object or objects chosen for analysis

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

With a fixed net force, a 4 kg system accelerates at 12 m/s².
Predict acceleration for a 8 kg system.
State a control needed for this comparison.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_{A \text{ on } B} = -F_{B \text{ on } A}$.
The two forces act on different objects at the same instant. Their accelerations can differ.

Worked reasoning: Two skaters push apart. Each feels 20 N from the other. If their masses are 40 and 80 kg, accelerations are 0.5 and 0.25 m/s² in opposite directions.

Your turn: Draw separate force diagrams for the horizontal interaction.
Explain why equal forces do not imply equal accelerations.

Sentence starter: My claim is … . The diagram/data show …, so … .



Skater A: 40 kg



Skater B: 80 kg

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Name both forces

A person presses a wall to the right. On the two-object model, draw the force the hand exerts on the wall and the force the wall exerts on the hand.
Label both forces with their source and receiving object.

Sentence starter: The … in my model represents … . Its label shows … .



Hand



Wall

Marzano check - circle: 1 2 3

04 Heavy and light carts

A heavy cart and a light cart push apart on a low-friction track. Add their interaction-force arrows to the model.
Compare force sizes and explain which cart accelerates more.

Sentence starter: Both … . They differ in … because … .



Light cart

Heavy cart

Marzano check - circle: 1 2 3

05 Balanced or a pair?

A book rests on a table. Earth pulls book down; table pushes book up. Are these the two forces of one interaction pair?
Explain by naming the object receiving each.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

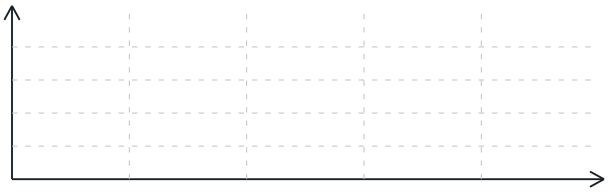
06 Review: connect an earlier idea

Start at 6 m, move to 16 m, then finish at 8 m.
Draw the route.
Find distance, displacement with direction and final position.
Label the model with the given positions and any prediction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I predict … . My model supports this because … .

Route



Position (m)

Marzano check - circle: ① ② ③

07 Exit check

A runner's foot pushes backward on the ground without slipping. Add the two interaction-force arrows to the model.
Name the partner force and the object it acts on.
Explain how the runner can accelerate forward.

Sentence starter: My claim is … . The diagram/data show …, so … .



Foot



Ground

Marzano check - circle: ① ② ③

08 STRETCH: test the model

Choose two masses that would give a 3: 1 acceleration ratio while the forces remain equal.
Explain which skater accelerates more.

Sentence starter: My claim is … . The diagram/data show …, so … .

Skater	Mass (kg)	Acceleration ratio
A	Choose: ____	3
B	Choose: ____	1

Marzano check - circle: ① ② ③